

Test Report: Host Health & Availability

1. Architectural Approach

- **Method:** Zabbix Agent 2 (Active Checks).
- **Security Justification:** Requires zero inbound firewall ports. Uses 100% Native Checks. The dangerous `system.run` feature was strictly avoided to prevent Remote Code Execution (RCE) vulnerabilities.

2. Pending Production Clarifications

Because this template acts as a Proof-of-Concept, generic standards were utilized to test the logic. Clarification is required on the following items before production rollout:

- **[OBS-F-004] Services:** The `sshd` service was targeted as a generic test service. Please provide the exact list of operational services to monitor.
- **[OBS-F-016] Critical Processes:** The `zabbix_agent2` process was targeted to test CPU/Mem limits. Please provide the exact application processes to monitor.

3. Metric Collection & Initial Error Discovery

- **Action:** Deployed the static template directly to the RHEL 9 VM via Ansible.
- **Result:** Below is the initial Latest Data ingestion, revealing expected security restrictions and parsing errors.

Save asApplyReset

Subfilter affects only filtered data

HOSTS

Rhel9-Not-Hardened 17

STATE

Normal 10Not supported 7

DATA

With dataWithout data

☐	Host	Name	Last check	Last value	Change	Tags	Actions	Info
☐	Rhel9-Not-Hardened	Available Memory	54s	93.0379 %	+0.000208 %		Graph	
☐	Rhel9-Not-Hardened	Boot Outcome Log					History	7
☐	Rhel9-Not-Hardened	CPU I/O Wait	55s	0.00417 %			Graph	
☐	Rhel9-Not-Hardened	CPU Utilisation	56s	0.06672 %			Graph	
☐	Rhel9-Not-Hardened	File Descriptor Usage Percentage					Graph	7
☐	Rhel9-Not-Hardened	Free Swap					History	7
☐	Rhel9-Not-Hardened	Global Max File Descriptors					History	7
☐	Rhel9-Not-Hardened	Kernel Errors Log					History	7
☐	Rhel9-Not-Hardened	Lifecycle Event Log					History	7
☐	Rhel9-Not-Hardened	Out of Memory Log					History	7
☐	Rhel9-Not-Hardened	Process CPU: Zabbix Agent	48s	0.7 %			Graph	
☐	Rhel9-Not-Hardened	Process Mem: Zabbix Agent	47s	0 B			Graph	
☐	Rhel9-Not-Hardened	Service Restarts: SSH					Graph	7
☐	Rhel9-Not-Hardened	Service Status: SSH	44s				Graph	
☐	Rhel9-Not-Hardened	System Uptime	57s	0:3:3:3:36	+0:0:0:1:0:0		Graph	
☐	Rhel9-Not-Hardened	Used File Descriptors					Graph	7
☐	Rhel9-Not-Hardened	Zabbix Agent Ping	58s	1			Graph	

0 selected

Displaying 17 of 17 found

Cannot open file "/var/log/messages": [13] Permission denied

Cannot evaluate function: Item "/[Rhel9-Not-Hardened]/vfs.file.contents[/proc/sys/fs/file-nr]" is not supported at "last(/vfs.file.contents[/proc/sys/fs/file-nr]) / last(/kernel.maxfiles)".

Cannot open file "/var/log/messages": [13] Permission denied

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Cannot get unit property: Unknown interface org.freedesktop.systemd1.Unit or property NRestarts.

Value of type "string" is not suitable for value type "Numeric (unsigned)". Value "1".

Initial Error Context:

- 1. **The Log Items (Permission Denied):** Linux blocked the agent from reading `/var/log/messages` because the agent runs securely as a non-root user.
- 2. **File Descriptors (Value of type string):** The native kernel read succeeded, but Zabbix's internal Regex preprocessing failed to isolate the integer.
- 3. **Service Restarts (Unknown property NRestarts):** Systemd only tracks `NRestarts` if a service has `Restart=on-failure` configured in its unit file (default `sshd` does not).

4. Remediation & Fixes Applied

To resolve the critical log monitoring requirement without compromising the Linux security boundary, I applied a strict Access Control List (ACL) exception.

- **The Fix:** Executed `setfacl -m u:zabbix:r /var/log/messages` on the RHEL OS.
- **Why it is secure:** This surgically granted read access to this single file without elevating the Zabbix agent to root privileges.

[Post-Remediation Verification]

(The log errors are resolved and natively collecting data)

Subfilter affects only filtered data

HOSTS

Rhel9-Not-Hardened 17

STATE

Normal 14 Not supported 3

DATA

With data Without data

☐	Host	Name	Last check	Last value	Change	Tags	Actions	Info
☐	Rhel9-Not-Hardened	Available Memory	49s	93.0465 %	+0.01006 %		Graph	
☐	Rhel9-Not-Hardened	Boot Outcome Log	36s	Sep 15 07:24:27 localhost systemd[1]: Failed to start Ne...			History	
☐	Rhel9-Not-Hardened	CPU I/O Wait	50s	0.00417 %	+0.00417 %		Graph	
☐	Rhel9-Not-Hardened	CPU Utilisation	51s	0.05837 %	-0.004188 %		Graph	
☐	Rhel9-Not-Hardened	File Descriptor Usage Percentage					Graph	
☐	Rhel9-Not-Hardened	Free Swap	48s	100 %			Graph	
☐	Rhel9-Not-Hardened	Global Max File Descriptors	3m 41s	9.2234E+18			Graph	
☐	Rhel9-Not-Hardened	Kernel Errors Log	34s	Sep 15 07:26:31 localhost NetworkManager[889]: <info...			History	
☐	Rhel9-Not-Hardened	Lifecycle Event Log	36s	Sep 15 07:22:53 localhost systemd[1]: Finished Restore...			History	
☐	Rhel9-Not-Hardened	Out of Memory Log					History	
☐	Rhel9-Not-Hardened	Process CPU: Zabbix Agent	43s	0.6 %			Graph	
☐	Rhel9-Not-Hardened	Process Mem: Zabbix Agent	42s	0 B			Graph	
☐	Rhel9-Not-Hardened	Service Restarts: SSH					Graph	
☐	Rhel9-Not-Hardened	Service Status: SSH	39s	active			History	
☐	Rhel9-Not-Hardened	System Uptime	52s	04:24:56	+00:01:00		Graph	
☐	Rhel9-Not-Hardened	Used File Descriptors					Graph	
☐	Rhel9-Not-Hardened	Zabbix Agent Ping	53s	1			Graph	

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Displaying 17 of 17 found

5. Threshold Trigger Verification

- **Action:** Manually stopped the `sshd` service on the RHEL node to test the threshold logic.
- **Result:** Zabbix instantly detected the `inactive` state and fired the designated `[OBS-F-004]` CRITICAL alarm.

[Trigger Alert Proof]

Show

Recent problems

Problems

History

Host groups

type here to search

Select

Hosts

Rhel9-Not-Hardened X

type here to search

Select

Triggers

type here to search

Select

Problem

Severity

☐ Not classified

☐ Warning

☐ High

☐ Information

☐ Average

☐ Disaster

Age less than

☐ 14

days

Host inventory

Type

Remove

Add

Tags

And/Or

Or

tag

Contains

value

Remove

Add

Show symptoms

☐

Show suppressed problems

☐

Acknowledgement status

All

Unacknowledged

Acknowledged

By me

Save as

Apply

Reset

0 selected

Displaying 1 of 1 found